

Information Technology in Business
BCC 110

Reference Book:
Management Information Systems MANAGING THE DIGITAL FIRM-17th Edition, Kenneth C. Laudon & Jane P. Laudon

Chapter 1
Information Systems in Global Business Today

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The Role of Information Systems in Business Today

- **How information systems are transforming business**
  - Emerging mobile digital platform
  - Growing business use of "big data"
  - Growth in cloud computing
- **Globalization opportunities**
  - Internet has drastically reduced costs of operating on global scale
  - Increases in foreign trade, outsourcing
  - Presents both challenges and opportunities

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The Role of Information Systems in Business Today

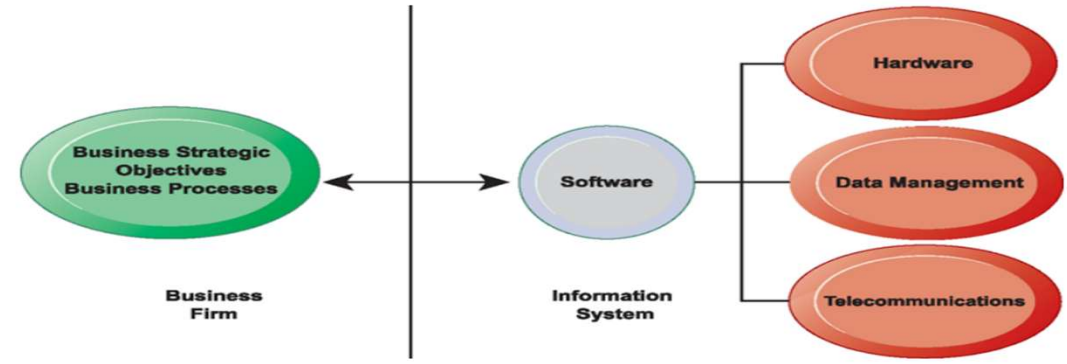
- **Digital firm:**
  - Nearly all of the significant business relationships with customers, suppliers & employees of an organization are digitally enabled and mediated.
  - Core business processes are accomplished through digital networks.
  - Key corporate assets (Intellectual property: copyright-patent, Core competencies, Financial & Human assets) are managed digitally.
  - Digital firms offer greater flexibility in organization and management.

- **Growing interdependence between:**
  - Ability to use information technology and
  - Ability to implement corporate strategies and achieve corporate goals

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The Interdependence between Organizations and Information Technology

FIGURE 1-2 THE INTERDEPENDENCE BETWEEN ORGANIZATIONS AND INFORMATION SYSTEMS



In contemporary systems there is a growing interdependence between a firm's information systems and its business capabilities. Changes in strategy, rules, and business processes increasingly require changes in hardware, software, databases, and telecommunications. Often, what the organization would like to do depends on what its systems will permit it to do.

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Information Systems

- **Information Systems:**
  - A set of interrelated components that collect, create, process, store, and distribute data and information to support an organization's operations, decision-making, and strategic objectives.
  - Support decision making, coordination, control, and strategic planning
  - Enable data-driven problem solving and innovation
  - Facilitate collaboration and communication across global networks
  - Provide digital solutions to dynamic business and societal challenges
  - **Component of IS: Hardware, Software, Data, Procedure, People.**

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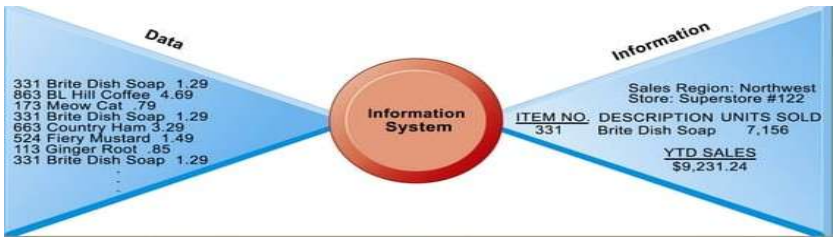
Data and Information

Data

- Streams of raw facts representing events such as business transactions.
- The raw material from which information is generated.
- Data appear in the form of text, numbers, figures or any combination of these.
- Must be processed to become useful.

Information

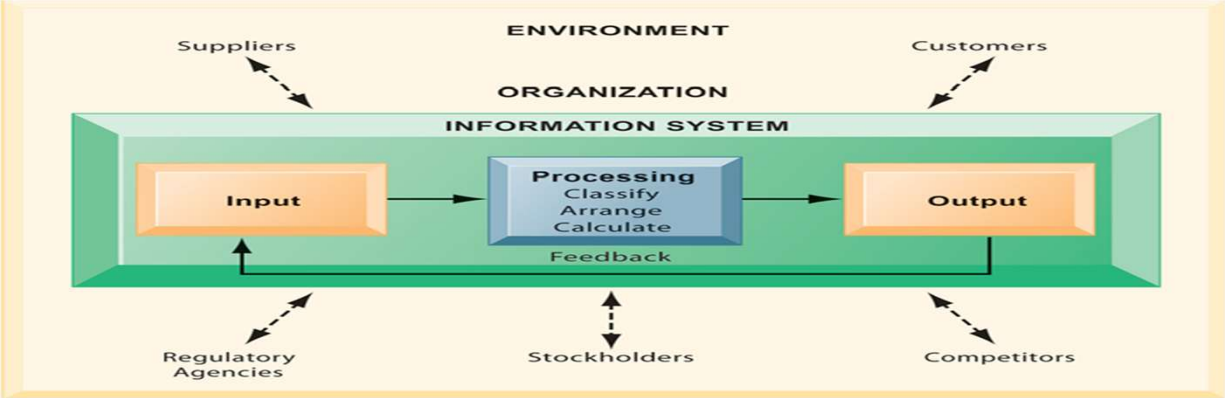
- Clusters of facts /data processed and converted into a form that are meaningful and useful to human beings in the processes such as making decisions.
- Facts, principles, knowledges, experience and intuition are applied to convert data into information.



In a way, *information is data that has been transformed into a more useful form*.

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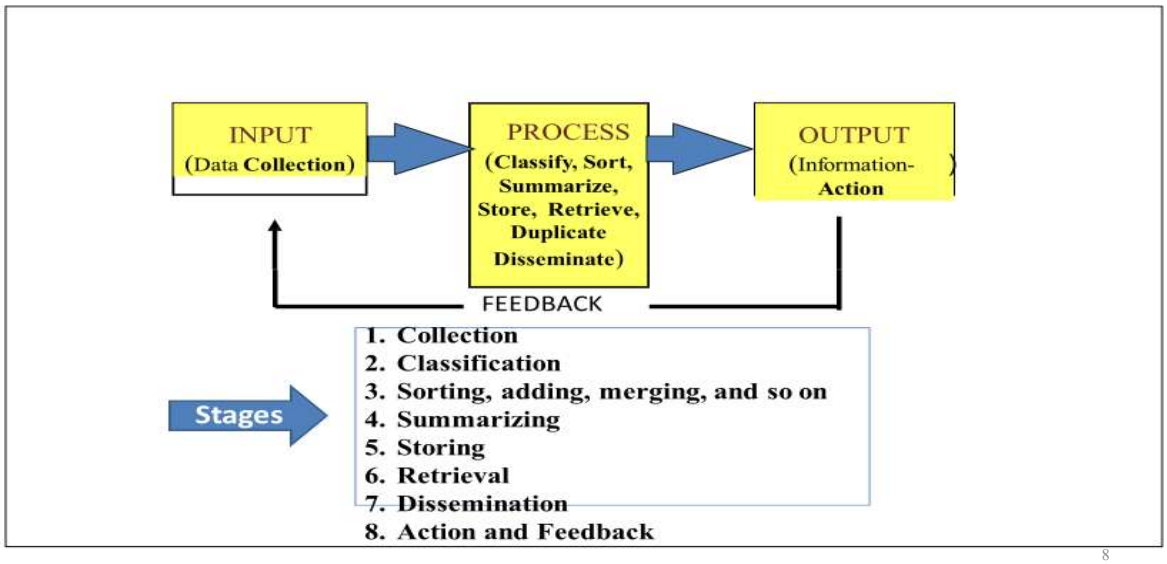
Activities/ Functions of an Information System



An information system contains information about an organization and its surrounding environment. Three basic activities—input, processing, and output—produce the information organizations need. Feedback is output returned to appropriate people or activities in the organization to evaluate and refine the input. Environmental actors, such as customers, suppliers, competitors, stockholders, and regulatory agencies, interact with the organization and its information systems.

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The Process of Converting Data into Information



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Characteristics of Good Information

- **Subjectivity:** Value and usefulness of information are highly subjective cause
  - What is information for one person may not be for other (change in stock price)
- **Relevancy :** Good only when it is relevant & meaningful to decision maker
- **Timeliness:** Must be delivered at the right time and the right place to the right one
- **Accuracy:** Must be free of errors at any stage
- **Correct Information Format:** In right format to be useful to the decision maker.
- **Completeness:** If decision maker satisfactorily solve the problem at hand using it.
- **Accessibility:** Is useless if it is not readily accessible to decision makers in desired format and when required.

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How Information Systems Are Transforming Business?

- Social networking tools being used by businesses to connect employees, customers, and managers.
- Internet advertising continues to grow at more than 20 percent per year
- New laws require businesses to store more data for longer periods
- Changes in business result in changes in jobs and careers

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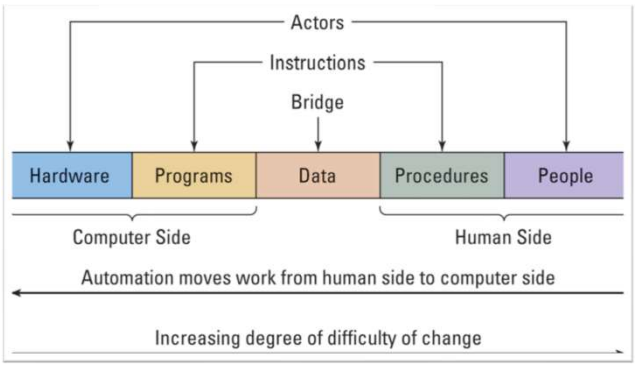
What's New in Management Information Systems?

- **I T Innovations**
  - Cloud computing, big data, Internet of Things
  - Mobile digital platform
  - AI and machine learning
- **New Business Models**
  - Online streaming music and video
  - On-demand e-commerce services
- **E-commerce Expansion**
  - E-commerce expands to nearly $1 trillion in 2018
  - Netflix now has more than 125 million U S subscribers
  - Online services now approach online retail in revenue
  - Online mobile advertising now larger than desktop
- **Management Changes**
  - Managers use social networks, collaboration tools
  - Business intelligence applications accelerate
  - Virtual meetings proliferate
- **Firms and Organizations Change**
  - More collaborative, less emphasis on hierarchy and structure
  - Greater emphasis on competencies and skills
  - Higher-speed/more accurate decision making based on data and analysis
  - More willingness to interact with consumers (social media)
  - Better understanding of the importance of I T

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Five-Component Framework of Information System

- The five-component framework can help guide your learning and thinking about information system both now and in the future. Automation occurs when a business process is moved to a computer to perform the business process.
- The Most Important Component-YOU



Core Components of Information Systems (IS):

1. Hardware: Physical devices (servers, computers, IoT sensors, mobile devices, etc.)
2. Software: Applications, operating systems, and cloud-based platforms that process data
3. Data: The lifeblood of IS; includes both structured and unstructured information
4. Procedures: Methods, policies, and rules governing data usage and system operations
5. People: Users, IT professionals, and decision-makers who interact with and manage the system

Information System	Information Technology
IS is a coordinated set of components and resources designed to collect, store, process, and distribute information. It typically involves the use of computer hardware and software, databases, and communication systems to help organizations manage their operations, interact with customers, and make informed decisions.	IT refers to the use of computers, networks, storage, and other physical devices, infrastructure, and processes to create, process, store, secure, and exchange all forms of electronic data. IT encompasses both the hardware and software necessary to manage and operate information systems
A Customer Relationship Management system. It helps businesses manage customer data, track interactions, and improve customer service. Salesforce is a popular CRM system that integrates various functionalities to support sales, marketing, and customer service.	**Cloud computing.** Amazon Web Services is a leading provider of cloud computing services, allowing businesses to run applications and store data without managing physical servers.

The difference between information systems and information technology is that information system incorporates the technology, people, and processes involved with information, Information technology is the design and implementation of information, or data, within the information system.

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Why Study Information Systems?

"It's the lack of understanding of the part of the upper management about how to oversee, budget, and come to grips with the real expense and value of the enterprise information infrastructure. And it's costing companies millions of dollars in cost overruns and futile attempts to improve productivity and roll out new products" - Data Communications

- Make you an educated end-user
- Contribute in the design of new systems (in your job)
- Increase organizational success
- Advance your career
- Help you meet the challenges in a global market where management decisions become more complex
- Learn how data and information is used by the organization (information systems literacy)

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Why Organizations Need Information Systems?

- Today, few if any medium-or large-sized companies can survive without computer and information systems. The world has become a small place, the competition is no longer limited by boundaries, the reasons why organization need information systems are...
- **Meeting Global Challenges:** Produce high quality goods and services that can compete in world markets
- **Capturing Opportunities In The Market Place:** Successful companies identify and take advantages of opportunities in market place through Strategic Information System(SIS)
- **Supporting Corporate Strategy:** Use 3 basic strategies : Cost strategy, product differentiation, market niche strategy.
- **Linking Departments Whose Functions Is Different:** accounting, finance, marketing, manufacturing, human resources are linked together.
- **Enhancing Worker Productivity:** Computer and IS have a significant impact on a company bottom line .
- **Increasing The Quality of Good and Services:** Using TQM for enhancing quality by top and middle managers.

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Globalization Challenges and Opportunities: A flattened World

- **Internet and global communications have greatly changed how and where business is done**
  - Drastic reduction of costs of operating and transacting on global scale
  - Competition for jobs, markets, resources, ideas
  - Growing interdependence of global economies
  - Requires new understandings of skills, markets, opportunities
  - Challenges yes, but opportunities as well
  - Over half of the revenue of S&P 500 US firms is generated off shore
  - Information systems enable globalization of commerce

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The Role of Information Systems in Business Today

- **How information systems are transforming business**
  - —Increase in wireless technology use, Web sites
  - —Increased business use of Web 2.0 technologies
    - —Cloud computing, mobile digital platform allow more
    - —distributed work, decision-making, and collaboration.
- **In the emerging, fully digital firm**
  - —Significant business relationships are digitally enabled and mediated
  - —Core business processes are accomplished through digital networks
  - —key corporate assets are managed digitally
- **Digital firms offer greater flexibility in organization and management**
  - —Time shifting, space shifting

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How IS transforms business today?

1. Creating globalization opportunities:

- Internet reduced costs of operating, on global scale for Customers and firms.
- Using internet service firms such as Google and e-bay, able to replicate business and services in multiple countries without higher investment.

2. Helps the emerging digital firm:

- The business relationships in a digital firm are digitally enabled and mediated and It's core business processes (set of logically related tasks and behaviors, like developing new product, generating and fulfilling and order, creating marketing plan, hiring an employee) are accomplished through digital networks.
- Key corporate assets (intellectual property, core competencies, financial and human assets) are manage through digital means.

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Firms invest heavily in IS to achieve 6 strategic business objectives (Why firms use IS?)

- Growing interdependence between ability to use information technology and ability to implement corporate strategies and achieve corporate goals.
- Business firms invest heavily in information systems to achieve six strategic business objectives:
  - ***Operational excellence (Improvement of efficiency to attain higher profitability)***
  - ***New products, services, and business models (Enabled by technology)***
  - ***Customer and supplier intimacy (Serving customers raises revenues and profits)***
  - ***Improved decision making (More accurate data leads to better decisions)***
  - ***Competitive advantage***
  - ***Survival***

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6 strategic Business Objectives

1. Operational Excellence:

- Information systems improve efficiency, streamline processes, and reduce costs, leading to higher profitability.
  - IS Improvement of efficiency of operation to attain higher profitability
  - IT tool to achieve greater efficiency and productivity
- *Toyota uses its Just-In-Time inventory system to reduce waste and improve manufacturing efficiency, which boosts profitability. FedEx uses advanced logistics and tracking systems to optimize its delivery network, improving efficiency and reducing costs. This leads to faster deliveries and higher profitability.*

2. New Products, Services, and Business Models:

- Technology enables firms to innovate, create new products/services, and develop new business models.
- *Netflix transformed the entertainment industry by using streaming technology, creating a new business model and disrupting traditional media consumption. Apple leveraged information systems to introduce the App Store, creating a new business model by allowing developers to sell apps directly to consumers. This transformed the mobile industry and created a new revenue stream.*

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6 strategic Business Objectives

3. Customer and Supplier Intimacy:

- Information systems enable businesses to understand and respond to customer needs and strengthen supplier relationships, increasing revenues and profits.
- *Amazon uses personalized recommendation systems based on customer browsing and purchase data, leading to increased sales and customer loyalty. It also strengthens supplier relationships by using efficient order management systems.*

4. Improved Decision Making:

- Access to accurate, real-time data helps businesses make better-informed decisions, leading to better outcomes.
- *Starbucks uses sales and customer data to optimize store locations and inventory, improving decision-making and profitability.*
- *Zara uses real-time sales data collected from its stores to quickly adjust inventory and design new clothing lines. This enables the company to make informed, data-driven decisions about what products to stock and produce.*

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6 strategic Business Objectives

5. Competitive Advantage:

- Information systems help firms gain a competitive edge through differentiation, cost leadership, or innovation.
- *Walmart uses its advanced supply chain management system to maintain lower prices and faster delivery, outperforming competitors.*
- *Nike uses advanced data analytics and digital platforms (like Nike Training Club and Nike Run Club) to engage customers, gather insights, and offer personalized experiences, which differentiates it from competitors in the athletic wear industry.*

6. Survival:

- Information systems are essential for keeping up with industry standards, complying with regulations, and responding to market demands.
- *Kodak failed to adapt to digital photography and the shift to online photo storage, ultimately leading to its downfall. Meanwhile, companies like Canon and Nikon adapted their business models, incorporating digital technology to survive and thrive in the digital photography market.*

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Solve the case

- Amazon continuously invests in advanced information systems to manage its vast supply chain, improve customer recommendations, and enhance operational efficiency. These systems help Amazon achieve faster deliveries, personalized services, and data-driven decisions.

- **Question:** Explain how Amazon's use of information systems helps the company achieve operational excellence, customer intimacy, improved decision-making, competitive advantage, and overall corporate success.

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Computer Literacy Vs. Information Literacy

Computer Literacy:

- The ability to efficiently use computers and digital technologies to perform tasks, solve problems, and communicate.
- Involves understanding **hardware, software, networks, and cloud systems**.
- Includes basic knowledge of **cybersecurity**, **AI tools**, and **data management**.
- Computers act as **extensions of human intelligence**, improving productivity and creativity.
- Focuses on using digital tools confidently and safely in both personal and professional contexts.

Information Literacy:

- The ability to **locate, evaluate, and use information effectively** through digital and information systems to support decision-making and gain a **competitive advantage**.
- Builds on **computer literacy** and adds **critical thinking** and **data literacy**.
- Requires understanding how **information systems** support organizational goals.
- Involves skills in **data analysis**, **digital communication**, and **ethical information use**.
- Enables individuals to **turn data into insights** for **innovation** and **strategic success**.

Dimensions of information systems

- To fully understand information systems, one must understand the broader organization, management and information technology. These are the **core perspectives** needed to fully understand how IS functions within an organization.
- ***Understanding of Information system dimensions is Information system literacy (technical).***
- ***Where computer literacy is focused on primarily on knowledge of IT.*** MIS try to achieve this broader information literacy.
- MIS deals with behavioral issues as well as technical issues surrounding the development, use, and impact of information systems used by managers and employees in the firm.



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Organizational dimension of information systems

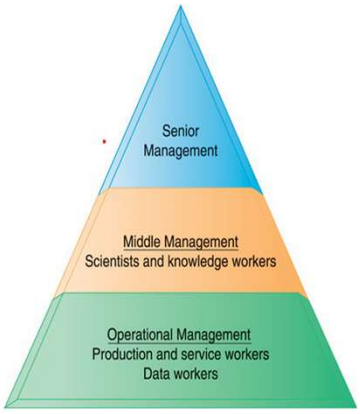
- **Information systems are an essential part of modern organizations**. In many industries, such as credit reporting, finance, and e-commerce, businesses cannot operate without them.
- An organization's key elements are its **people, structure, business processes, politics, and culture**. These factors influence how information systems are used and managed.
- Most organizations have a **hierarchical structure** with defined roles and responsibilities.
  - **Upper levels:** managers, professionals, and technical staff handle strategic decisions.
  - **Lower levels:** operational employees perform routine activities and service delivery.
- Today, **digital technologies** link all levels of the organization, improving communication, efficiency, and decision-making.

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Organizational dimension of IS

- **Senior management** makes long-range strategic decisions about products and services as well as ensures financial performance of the firm.
- **Middle management** carries out the programs and plans of senior management and operational management is responsible for monitoring the daily activities of the business.
- **Knowledge workers**, such as engineers, scientists, or architects, design products or services and create new knowledge for the firm, whereas **data workers**, such as secretaries or clerks, assist with scheduling and communications at all levels of the firm.
- **Production or service workers** actually produce the product and deliver the service.

FIGURE 1-6 LEVELS IN A FIRM



Business organizations are hierarchies consisting of three principal levels: senior management, middle management, and operational management. Information systems serve each of these levels. Scientists and knowledge workers often work with middle management.

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Organizational dimension of IS

- Experts are employed and trained for different business functions. The major business functions, or specialized tasks performed by business organizations, consist of sales and marketing, manufacturing and production, finance and accounting, and human resources.

TABLE 1-2 MAJOR BUSINESS FUNCTIONS

FUNCTION	PURPOSE
Sales and marketing	Selling the organization's products and services
Manufacturing and production	Producing and delivering products and services
Finance and accounting	Managing the organization's financial assets and maintaining the organization's financial records
Human resources	Attracting, developing, and maintaining the organization's labor force; maintaining employee records

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Organizational dimension of information systems

- An organization coordinates work through its hierarchy and through its business processes, which are logically related tasks and behaviors for accomplishing work. Developing a new product, fulfilling an order, and hiring a new employee are examples of business processes. Most organizations' business processes include formal rules that have been developed over a long time for accomplishing tasks. These rules guide employees in a variety of procedures, from writing an invoice to responding to customer complaints.
- Some of these business processes have been written down, but others are informal work practices, such as a requirement to return telephone calls from co-workers or customers, that are not formally documented. Information systems automate many business processes. *For instance, how a customer receives credit or how a customer is billed is often determined by an information system that incorporates a set of formal business processes.*
- Each organization has a unique culture, or fundamental set of assumptions, values, and ways of doing things, that has been accepted by most of its members.
- Different levels and specialties in an organization create different interests and points of view. These views often conflict over how the company should be run and how resources and rewards should be distributed. Conflict is the basis for organizational politics. Information systems come out of this cauldron of differing perspectives, conflicts, compromises, and agreements that are a natural part of all organizations.

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Management dimension of IS

- Management's job is to make sense out of the many situations faced by organizations, make decisions, and formulate action plans to solve organizational problems.
- **Managers perceive business challenges in the environment; they set the organizational strategy for responding to those challenges; and they allocate the human and financial resources to coordinate the work and achieve success.**
- Managers set organizational strategy for responding to business challenges. In addition, managers must act creatively and do more than what already exists:
  - *Creation of new products and services.*
  - *Occasionally re-creating the organization.*

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Technology dimension of IS

- Information technology is one of many tools managers use to cope with change. Information Technology is:
  - **Hardware:** physical components use for inputs
  - **Software:** preprogrammed instructions to control Hardware
  - **Data management technology:** Software that organizes data
  - **Networking and telecommunications technology:** *Networking: Hardware + Software+ link computers*
  - **Internet:** "global network of networks" the Universal
    - **Intranet:** Internal corporate networks based on internet technology
    - **Extranet:** Private intranets extended to authorized users outside the
  - **WWW**: "World Wide Web"- service to storing, retrieving, formatting, displaying information in a page .
  - **IT infrastructure**: platform that the firm can built on its specific information system

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Solve the case

Delta Bank recently launched a new cybersecurity management system to strengthen its data protection and regulatory compliance. The system uses AI-driven threat detection, multi-factor authentication, and cloud-based monitoring to safeguard customer information. However, several challenges have emerged after implementation:

- Many employees find the new login and verification process complicated, leading to frequent lockouts.
- Some departments continue using outdated data-sharing methods, creating security gaps.
- The IT team is overwhelmed managing alerts and system updates due to limited automation and training.

Question:

- Analyze the case using the three dimensions of information systems — Technical, Organizational, and Management.
- Identify the key issues for each dimension and suggest practical solutions to improve the system's effectiveness and adoption.

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Business perspective on IS

•Managers and business firms invest in information technology and systems because they provide real economic value to the business. The decision to build or maintain an information system assumes that the returns on this investment will be superior to other investments in buildings, machines, or other assets.

•**From a business perspective; an information system is an important instrument for creating value for the firm. Information systems enable the firm to increase its revenue or decrease its costs by providing information that helps managers make better decisions or that improves the execution of business processes.**

- Investments in information technology will result in superior returns:
  - Productivity increases
  - Revenue increases
  - Superior long-term strategic positioning
  - Data driven decision-making
  - Competitive advantage

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Business perspective on IS

- **An information system is not just technology—it is a solution that aligns technology with business strategies. It helps organizations operate efficiently, make informed decisions, manage resources, and achieve competitive advantages by collecting, processing, and analyzing data.**

Key Elements of the Business Perspective on IS

- **Organizational Goals**: IS supports organizational objectives, such as increasing revenue, reducing costs, or improving customer satisfaction.
- **Business Processes**: IS improves workflows by automating tasks, reducing errors, and ensuring consistency.
- **Decision-Making**: IS provides data and analytics for making informed strategic decisions.
- **Competitive Advantage**: IS can differentiate a business by offering unique products/services, reducing operational costs, or enabling better customer experiences.

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Business perspective on IS

Example: Amazon's Use of Information Systems

- **Goal Alignment**: Amazon's goal is to provide fast and reliable customer service. It uses its information systems to ensure efficient supply chain management and personalized recommendations.
- **Enhanced Business Processes**: Amazon's IS, such as its **inventory management system**, tracks stock levels in real time and ensures products are always available.
- **Improved Decision-Making**: Amazon's **data analytics systems** analyze customer behavior to recommend products and adjust inventory levels based on demand predictions.
- **Gaining Competitive Advantage**: Amazon's **delivery systems**, powered by IS, use optimized routing to ensure faster delivery than competitors, creating a strong competitive edge.

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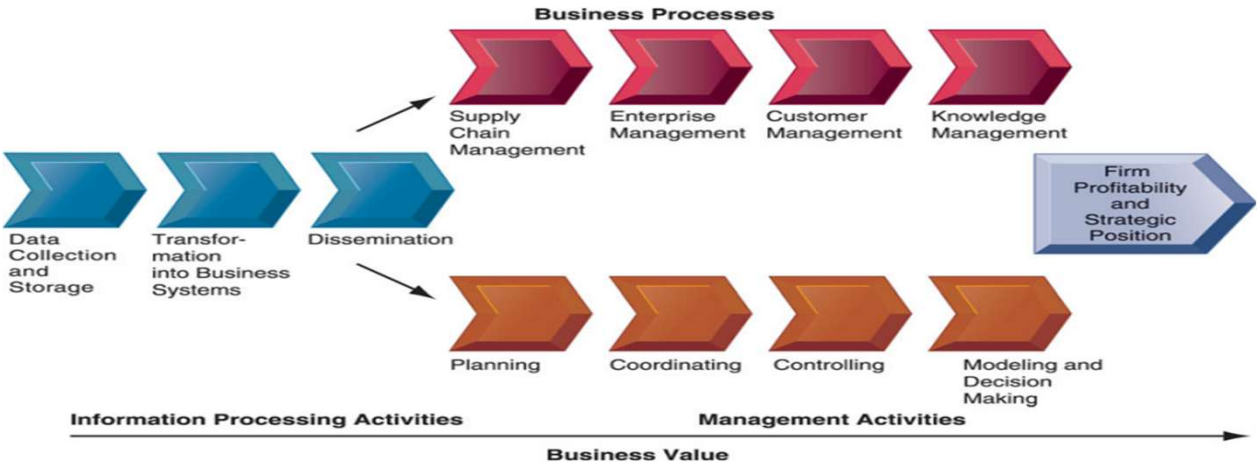
The Business Information Value Chain

- Every business has an information value chain, illustrated in Figure 1-7, in which **raw information is systematically acquired and then transformed through various stages that add value to that information.**
- The value of an information system to a business, as well as the decision to invest in any new information system, is, in large part, determined by the extent to which the system will lead to better management decisions, more efficient business.
- **Business information value chain**
  - ***Raw data acquired and transformed through stages that add value to that information.***
  - ***Value of information system determined in part by extent to which it leads to better decisions, greater efficiency, and higher profits.***
- Business perspective: Calls attention to organizational and managerial nature of information systems.

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The Business Information Value Chain

FIGURE 1-7 THE BUSINESS INFORMATION VALUE CHAIN



From a business perspective, information systems are part of a series of value-adding activities for acquiring, transforming, and distributing information that managers can use to improve decision making, enhance organizational performance, and, ultimately, increase firm profitability.

The Business Information Value Chain

- The **Business Information Value Chain** describes how information systems (IS) create value for businesses by transforming data into useful information and insights, which in turn help organizations improve processes, decision-making, and ultimately achieve business goals.

- **Netflix and the Business Information Value Chain:** Netflix is a well-known example of a company that effectively uses the **Business Information Value Chain** to provide value to its customers and gain a competitive edge in the streaming industry.

1. Data Collection: Netflix collects massive amounts of raw data, including:
- •What users watch (movies, series, genres).
- •When and how long users watch.
- •User interactions, such as pausing, rewinding, or skipping.
- •Ratings and reviews provided by users.
- •Device usage data (TV, phone, tablet, etc.).

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The Business Information Value Chain

2. Data Processing

- The raw data is processed and organized using advanced data analytics tools and algorithms. Key activities include:
- Categorizing viewing habits by genre, time of day, and demographics.
- Grouping users into clusters based on their preferences (e.g., "comedy lovers" or "true crime fans").
- Filtering out irrelevant or noisy data to focus on actionable insights.

3. Information Generation

- Processed data is turned into actionable insights, such as:
- Predicting what type of content a user is likely to enjoy.
- Identifying trends, such as the rising popularity of a specific genre.
- Determining which original content to produce or license (e.g., Netflix Originals).

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The Business Information Value Chain

5. Business Processes

- These decisions improve business operations, such as:
- Streamlining the production of Netflix Originals to align with viewer preferences.
- Optimizing the streaming experience by preloading likely-to-be-watched content for users.

6. Business Value Creation

- The result is significant value creation for Netflix:
- **Customer Satisfaction**: Personalized recommendations keep users engaged, increasing retention rates.
- **Revenue Growth**: Increased user satisfaction drives subscriptions and renewals.
- **Competitive Advantage**: Netflix leverages its data-driven insights to stay ahead of competitors like Hulu and Disney+.
- **Cost Efficiency**: By producing content that aligns with user preferences, Netflix avoids costly failures.

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Complementary Assets: Organizational Capital And The Right Business Model

- Information technology investments alone cannot make organizations and managers more effective unless they are accompanied by supportive values, structures, and behavior patterns in the organization and other complementary assets. Some firms fail to adopt the right business model that suits the new technology, or seek to preserve an old business model that is doomed by new technology. *For instance, recording label companies refused to change their old business model, which was based on physical music stores for distribution rather than adopt a new online distribution model. As a result, online legal music sales are dominated not by record companies but by a technology company called Apple Computer.*
- Complementary assets are those assets required to derive value from a primary investment (Teece, 1988). **It refer to resources, capabilities, or organizational elements that must be present for a primary investment—such as information technology (IT)—to deliver maximum value. Without these complementary assets, the primary investment may not yield significant benefits.** *For instance, to realize value from automobiles requires substantial complementary investments in highways, roads, gasoline stations, repair facilities, and a legal regulatory structure to set standards and control drivers.*

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Complementary Assets: Organizational Capital And The Right Business Model

TABLE 1-3 COMPLEMENTARY SOCIAL, MANAGERIAL, AND ORGANIZATIONAL ASSETS REQUIRED TO OPTIMIZE RETURNS FROM INFORMATION TECHNOLOGY INVESTMENTS

Organizational assets	Supportive organizational culture that values efficiency and effectiveness Appropriate business model Efficient business processes Decentralized authority Distributed decision-making rights Strong IS development team
Managerial assets	Strong senior management support for technology investment and change Incentives for management innovation Teamwork and collaborative work environments Training programs to enhance management decision skills Management culture that values flexibility and knowledge-based decision making.
Social assets	The Internet and telecommunications infrastructure IT-enriched educational programs raising labor force computer literacy Standards (both government and private sector) Laws and regulations creating fair, stable market environments Technology and service firms in adjacent markets to assist implementation

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Types of Complementary Assets

1. Organizational Assets: These are internal structures and systems that allow technology to work effectively within the company.

- Supportive culture valuing efficiency and effectiveness
- Efficient business processes
- Decentralized authority
- Distributed decision-making rights
- Strong IS (Information System) development team

- Toyota's success with its **Toyota Production System (TPS)** isn't only because of advanced IT systems but also because of its **efficient business processes** and a **culture of continuous improvement (Kaizen)**.
  - Employees at all levels are encouraged to identify inefficiencies and make improvements (distributed decision-making).
  - The organization's culture values efficiency and teamwork — this organizational support ensures IT investments (like production tracking systems) are fully utilized.

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Types of Complementary Assets

2. Managerial Assets: These involve leadership capabilities and management practices that support and guide technology use.

- Senior management support for technology investment
- Incentives for innovation
- Teamwork and collaboration
- Training programs for decision-making
- Flexible, knowledge-based management culture

- Amazon's leaders strongly support **data-driven decision-making**.
  - They encouraged managers to use IT analytics to improve operations.
  - Amazon's management culture promotes innovation and flexibility, allowing teams to experiment with new technology (like AI or robotics).
  - Continuous **training programs** and management support enable Amazon to implement technology effectively.

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Types of Complementary Assets

3. Social Assets: These exist outside the organization, in society or the industry and help IT work effectively.

- Internet and telecom infrastructure
- IT-enriched education programs
- Standards and regulations
- Fair, stable markets
- Support from technology and service firms

- *Bangladesh's Mobile Banking (bKash). Its* success depends not only on its mobile app but also on **social assets** such as:
  - The availability of **mobile network infrastructure** and **internet coverage**.
  - **Government regulations** supporting digital finance.
  - **Public computer literacy** and trust in mobile technology. Without these societal supports, bKash could not have scaled so widely.

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Contemporary Approaches to Information Systems

- The study of information systems is a multidisciplinary field. No single theory or perspective dominates. In general, the field can be divided into **technical and behavioral approaches.**
- Information systems are sociotechnical systems. Though they are composed of machines, devices, and "hard" physical technology, they require substantial social, organizational, and intellectual investments to make them work properly.
- Contemporary approaches to information systems recognize that technology alone is not enough. By combining technical expertise with behavioral insights, organizations can build systems that are both efficient and user-friendly.
- *For example, Amazon's predictive inventory management (technical) paired with its emphasis on customer satisfaction (behavioral) showcases the synergy of these approaches in a modern business context.*

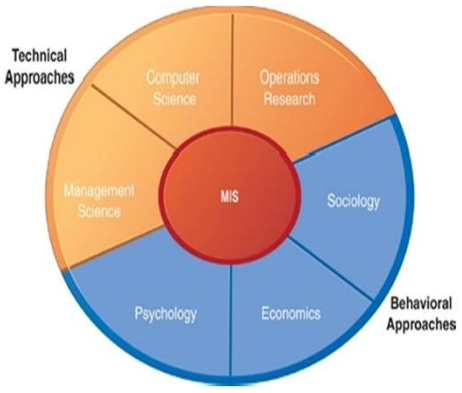


Figure 1-9 illustrates the major disciplines that contribute problems, issues, and solutions in the study of information systems

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Contemporary Approaches to Information Systems

1. Technical Approaches:

- These focus on the formal models and capabilities of information systems, emphasizing technology and efficiency.
- The technical approach to information systems emphasizes mathematically based models to study information systems, as well as the physical technology and formal capabilities of these systems.
- The disciplines that contribute to the technical approach are **computer science, management science, and operations research.**

a) Management Science Approach: Developing mathematical and statistical models to optimize decision-making and operations.

- ***Amazon****: Uses machine learning algorithms (a management science technique) to optimize its inventory management. Predictive analytics ensures warehouses stock the right products based on customer demand, minimizing overstock and understock situations.*

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Contemporary Approaches to Information Systems

b) Computer Science Approach: The study of computing, software, and hardware to design efficient information systems.

- ***Google Search Engine****: Google's information retrieval system is built using advanced computer science principles, such as algorithms and data structures, ensuring fast and accurate search results.*

c) Operations Research Approach: Applying mathematical techniques to solve organizational problems such as logistics, scheduling, and resource allocation.

- ***FedEx****: Uses operations research to optimize its delivery routes. By analyzing factors like traffic, weather, and distance, FedEx ensures faster and more cost-efficient deliveries.*

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Contemporary Approaches to Information Systems

2. Behavioral Approaches

- These focus on the human and organizational aspects of information systems, ensuring that systems are user-friendly, ethical, and aligned with business goals.
- An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems.
- Issues such as strategic business integration, design, implementation, utilization, and management cannot be explored usefully with the models used in the technical approach. Other behavioral disciplines contribute important concepts and methods.

a) Sociology Approach: Studying the impact of information systems on individuals, groups, and organizations.

- ***Workplace Collaboration Tools****: Microsoft Teams is designed to foster collaboration among employees. Sociological studies of team dynamics help refine features like video calls, shared documents, and chat threads.*

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Contemporary Approaches to Information Systems

b) Psychology Approach: Examining how individuals interact with systems to design user-friendly interfaces and reduce cognitive load.

- ***Apple's iPhone Interface****: Apple's intuitive design principles, such as gesture-based navigation and minimalistic layouts, are rooted in understanding user psychology to improve user experience.*

c) Economics Approach: Analyzing the cost-benefit aspects of information systems to ensure value creation.

- ***Netflix****: Uses economic models to assess the cost of content delivery versus the value of customer satisfaction. Investments in streaming infrastructure, such as high-quality servers, are based on economic justifications.*

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Contemporary Approaches to Information Systems

3. Socio-Technical Systems Approach

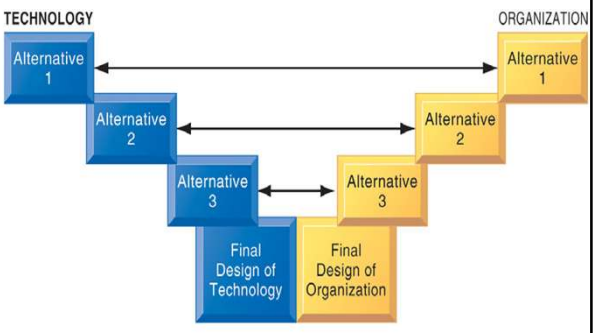
- This combines both technical and behavioral perspectives, emphasizing that organizations need to balance both aspects to ensure the success of an information system.
- A **sociotechnical system** is an approach to understanding the interaction between people (social factors) and technology (technical factors) in an organization. It emphasizes that both social and technical elements must be considered together to achieve optimal performance and effectiveness.
  - ***Tesla's Autopilot System***:
    - ***Technical Component***: *Sophisticated AI algorithms, sensors, and cameras.*
    - ***Behavioral Component***: *Ensuring drivers are aware of system limitations and providing interfaces that encourage safe usage (e.g., hands-on-wheel alerts).*

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Sociotechnical Systems

- Sociotechnical view of systems is optimal organizational performance is achieved by jointly optimizing both the social and technical systems used in production.
- Adopting a sociotechnical systems perspective helps to avoid a purely technological approach to information systems.
- For instance, the fact that information technology is rapidly declining in cost and growing in power does not necessarily or easily translate into productivity enhancement or bottom-line profits.
- Likewise, the fact that a firm has recently introduced new business procedures and processes does not necessarily mean employees will be more productive in the absence of investments in new information systems to enable those processes.

FIGURE 1-10 A SOCIOTECHNICAL PERSPECTIVE ON INFORMATION SYSTEMS



In a sociotechnical perspective, the performance of a system is optimized when both the technology and the organization mutually adjust to one another until a satisfactory fit is obtained.

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Sociotechnical Systems: Example: Uber Ride-Sharing Platform

1. **Technical System**:
   1. Uber's app, GPS, payment system, and backend servers form the technical infrastructure. The app matches riders with drivers, calculates fares, and processes payments in real-time.
2. **Social System**:
   1. Uber drivers and passengers interact through the app. Drivers use the app to receive ride requests, navigate to passengers' locations, and complete the journey. Passengers use the app to book rides, track drivers, and make payments.
   2. The company also has a customer support team to assist with issues that arise during rides, ensuring a smooth experience for both drivers and passengers.

- **Sociotechnical Integration**:
  - Uber's success depends on both the technology (the app's efficiency, real-time updates, payment processing) and the social interaction between drivers and passengers.
  - Problems like drivers not being able to navigate the app or disputes between drivers and passengers affect the system's overall performance, demonstrating the need for both well-functioning technology and effective human interaction.

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Potential Risks of Information system

- **"Deskilling "of workers**: While Is can increase operational efficiency and improve profits, sometimes it also the root cause of work-force reduction.
- **Information overload:** As IS generate reports easily and quickly, however this improvement can sometimes lead to overload /excessive of information.
- **Employee Mistrust:** Companies use to monitor the activities of employees which may lead to cause of mistrust and dislike .
- **Increased Competitive Pressure:** As growing development of IS , large companies with greater investment in IS pushed out the small companies in competition.
- **Misinvestment with IS:** Investment in inappropriate technology or poor applications of good technologies result in poor returns and decrease the value of Is and thus generate loss of the firm.
- **Challenges in Developing IS:** The most difficult task and challenging because of time constraint and budget.
- **Security Breaches:** The risk of securing the assets from theft and hacking.

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Management Information Systems

- **MIS as a field of study:** *An interdisciplinary field that is influenced by computer science, political science, psychology, operation research, linguistics, sociology and organizational theory and behavior.*
- **MIS as an Information system:** *MIS is a broad class of systems that provides information to facilitate organizational decision making.*
- MIS is the development and use of information systems that help businesses achieve their goals and objectives. There are three key elements:
    - *Components of an information systems*
    - *Development and use of information systems*
    - *Achieving business goals and objectives*
- MIS deals with behavioral issues as well as technical issues used by managers and employees of the firm.
- **Management information system (MIS) is a system that provides information needed to manage organizations effectively. Management information systems involve three primary resources: technology, information, and people.**

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Management Information Systems

- **Management Information System (MIS)** is a system used to collect, process, store, and disseminate information to help in managerial decision-making. It integrates data from different sources, processes it, and provides meaningful insights to support decision-making at various levels of an organization. A real-life example of MIS is **Walmart's inventory and sales system**. Here's how it works:

1. **Data Collection**: Walmart tracks sales, inventory levels, and customer preferences through sensors, barcode scanners, and loyalty programs.
2. **Data Processing**: The system analyzes sales trends, inventory data, and customer behavior to identify patterns.
3. **Information Storage**: Processed data is stored in a centralized database, accessible to managers and executives.
4. **Information Dissemination**: MIS generates reports on sales, stock levels, and market trends for decision-makers.
5. **Decision Support**: Managers use the data for inventory replenishment, sales forecasting, and product placement to optimize operations.

- Walmart's MIS helps streamline operations, reduce costs, and make informed, data-driven decisions.

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MIS Trends in Business

- **AI Integration**:
  - Boosts decision-making, automates tasks, and personalizes customer experiences.
  - **Amazon** uses AI to optimize inventory and recommend products, increasing sales and reducing costs.
- **Cloud Computing**:
  - Reduces infrastructure costs, enables scalability, and improves collaboration.
  - **Salesforce** integrates CRM functions to enhance customer relationships and productivity.
- **Big Data Analytics**:
  - Provides insights into customer behavior and operational performance for data-driven decisions.
  - **Walmart** uses big data to optimize inventory and improve supply chain efficiency.
- **Blockchain**:
  - Enhances data security and transparency, reducing fraud and increasing trust.
  - **IBM Food Trust** tracks food products via blockchain for transparency and safety.
- **Real-Time Data Processing**:
  - Enables quick responses to market changes and operational disruptions.
  - **Uber** uses real-time data to optimize ride matching and routes.
- **Mobile MIS**:
  - Provides on-the-go access to data for faster decision-making.
  - **Microsoft Power BI** enables executives to access business insights from mobile devices.
- These innovations enhance business efficiency, improve customer experience, and enable quicker, data-driven decision-making.

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